in Figure 2.14. Due to this symmetrical distribution, the shielding of one electron on the other is relatively small and hence the electrons are attracted more strongly by the nucleus and it increases the stability.

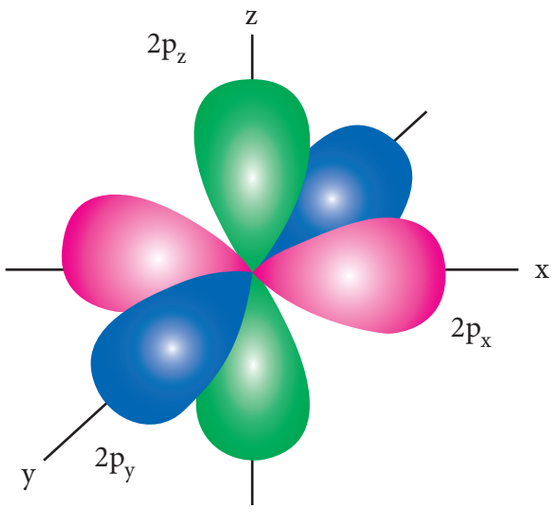
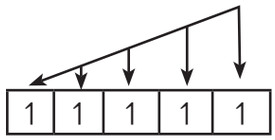


Figure 2.14 Shape of the degenerate p orbitals.

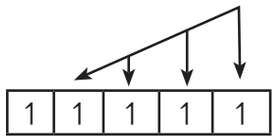
Exchange energy:

If two or more electrons with the same spin are present in degenerate orbitals, there is a possibility for exchanging their positions. During exchange process the energy is released and the released energy is called exchange energy. If more number of exchanges are possible, more exchange energy is released. More number of exchanges are possible only in case of half filled and fully filled configurations.

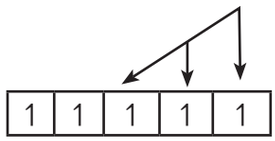
For example, in chromium the electronic configuration is $[Ar]3d^5\ 4s^1$. The 3d orbital is half filled and there are ten possible exchanges as shown in Figure 2.15. On the other hand only six exchanges are possible for $[Ar]3d^4\ 4s^2$ configuration. Hence, exchange energy for the half filled configuration is more. This increases the stability of half filled 3d orbitals.



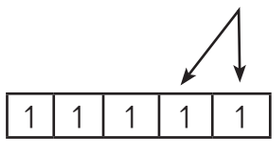
4 Exchanges by electrons



3 Exchanges by electrons



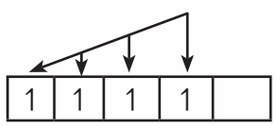
2 Exchanges by electrons



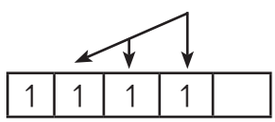
1 Exchange by electrons

Overall electron exchanges (4+3+2+1 = 10)

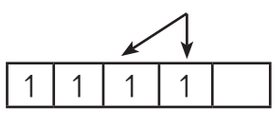
Figure 2.15 (a)



3 Exchanges by electrons



2 Exchanges by electrons



1 Exchange by electrons

Overall electron exchanges (3+2+1 = 6)

Figure 2.15 (b)

Figure 2.15 Possible exchanges in chromium d orbitals: a) for d^5 configuration b) for d^4 configuration

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